

Name: _____

Date: _____

LEARNING INTENTION	identify error, compare findings, draw conclusions and communicate for an audience
KEY IDEA	A conclusion answers the question using evidence and acknowledges limits. Evaluation compares methods, identifies error and proposes a specific next investigation; communication suits purpose and audience.
WORKED MODEL	Use the cycle: question -> prediction -> safe method -> precise data -> representation -> evidence-based conclusion -> evaluation -> communication.

SUCCESS CRITERIA - I CAN☐ apply the inquiry skill ☐ use evidence ☐ evaluate quality ☐ explain a next step**A. NOTICE AND IDENTIFY**

1. Match claim, evidence, reasoning, limitation and conclusion.

2. Distinguish random error, systematic error and procedural mistake.

3. Explain why comparing methods can reveal reliability problems.

B. APPLY AND EXPLAIN

4. Write a conclusion using numerical evidence from the scenario.

5. Identify one source of error and its likely direction/effect.

6. Pose a specific follow-up question that extends rather than repeats the investigation.

C. INDEPENDENT FLUENCY - CHECK AND CREATE

7. Adapt the same finding for a scientist, school community and younger audience.

8. Check purpose, vocabulary, visual evidence and degree of certainty.

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D. APPLY THE SKILL IN CONTEXT

9. Scenario: compare insulation materials using repeated cooling data Group A finds 14%, Group B 16%, Group C 29%. Compare findings.

10. Decide whether the evidence supports a common conclusion and investigate the outlier.

11. Write a reasoned conclusion with limitation and next step.

E. REASON, EVALUATE AND TRANSFER

12. Evaluate a conclusion that says 'proved' after one small trial.

13. Create a 100-word report and a four-point visual summary for different audiences.

F. STUDENT REVIEW - PROVE YOUR UNDERSTANDING

[] I can identify error, compare findings, draw conclusions and communicate for an audience.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult decision was _____ because _____

One correction or improvement I made: _____

My next practice step: _____

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Accept equivalent responses when the student applies the targeted skill accurately and justifies the choice.

COMPLETE ANSWER KEY

1. Claim answers; evidence=data; reasoning links; limitation bounds; conclusion synthesises.
2. Random varies; systematic shifts consistently; mistake breaches method.
3. Differences expose controls, instruments, samples or interpretation.
4. Conclusion must answer question and cite values.
5. State specific source and plausible effect, not 'human error'.
6. Follow-up changes scope or tests a new relationship.
- 7-8. Accuracy stays constant while language/detail/visuals change for audience.
9. A/B are similar; C requires method/data investigation.
10. Tentative common pattern may be supported after checking C.
11. Include evidence, uncertainty, limit and precise next step.
12. Replace proof claim with proportional, tentative wording.
13. Both formats must preserve evidence and uncertainty.

TEACHER CHECK AND NEXT STEP

[] Uses accurate science ideas and evidence.

[] Explains a decision rather than listing an answer.

Next step: model one difficult item, complete one together, then transfer the skill to a new investigation.

COMMON MISCONCEPTION / WATCH OUT

Do not write 'human error' without naming the specific cause and effect.

A conclusion must answer the question and stay proportional to the evidence.

EXTENSION AND RE-TEACH

Re-teach: model one item aloud, complete one together, then ask the student to explain an independent correction.

Extension: transfer the skill to a short authentic text and justify two deliberate language choices with quoted evidence.